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DEPARTMENT OF CSE(AI&ML)

Real-Time Research Project

on

Visual Study of Price Volatility Using Candlestick Charts and Return Distribution

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Abstract

In recent trading data, the market opened at 15,230, reached a high of 15,480, dropped to 15,120, and closed at 15,350 with a trading volume of 2.3 million shares.

These changes show that the stock market is very dynamic and volatile, which makes analysis difficult.

This project focuses on analyzing stock market behavior using a simple and visual approach.

The dataset includes details like Date, Open, High, Low, Close, and Volume, which help us understand daily trading activity.

We use candlestick charts to clearly show price movements and identify trends like upward (bullish) and downward (bearish) movements.

We also calculate daily returns from closing prices and study their distribution to understand how much the prices change over time.

A dashboard is created to combine these visualizations, making it easy for users to explore stock trends and volatility in one place.

Unlike traditional methods that use only numbers, this project combines visual charts and return analysis, making complex data easier to understand.

The main goal is to provide a simple and effective way to analyze stock price movements and volatility, helping users make better decisions.

Keywords : Stock Market Behavior, Price Volatility, Candlestick Charts, Return Analysis, Return Distribution, Financial Data, Data Visualization, Time Series Data, Trend Analysis, Interactive Dashboard, Market Fluctuations, Trading Volume

Introduction

- Financial markets keep changing due to different economic, political, and market factors
- To make good decisions, it is important to understand how prices move
- First, price movements are analyzed using candlestick charts
- These charts show open, high, low, and close prices
- They help in identifying trends and patterns in the market
- Next, price volatility is studied to understand how much prices change over time
- This helps in measuring market risk
- Then, return distribution is used to see how price changes are spread over time
- It helps in understanding frequency and variation of returns
- Finally, combining these methods gives a clear understanding of market behavior
- It helps users analyze risk and make better financial decisions

Motivation

- Price volatility is important because stock markets are uncertain and keep changing
- Market changes happen due to economic factors, investor behavior, and global events
- Understanding volatility helps in measuring risk and finding investment opportunities
- Numerical data alone is difficult to understand and does not clearly show market behavior
- Visual tools help in simplifying complex financial data
- Candlestick charts show price movements clearly
- They help in identifying trends, reversals, and market direction
- They also represent the behavior of buyers and sellers
- Return distribution dashboards show how prices change over time
- They help in understanding frequency and variation of returns

Problem Definition

- Financial markets are complex and unpredictable
- Prices change due to economic conditions, investor behavior, and global events
- These changes make it difficult to clearly understand price volatility
- Traditional methods use numerical and statistical data
- But they are hard to understand and do not give clear visual insight
- Candlestick charts are useful but need proper knowledge to identify trends and patterns
- Return distribution shows price changes and risk
- But without visualization, it is difficult to interpret

Literature Survey

Author(s)	Year	Model/Method	Approach	Results
Chen et al. [1]	2022	SSA-CBPGRU	Uses candlestick chart features combined with Sparrow Search Algorithm (SSA) for hyperparameter optimization of a BiGRU model	Improved prediction performance: ↓31.13% MAPE, ↓24.92% RMSE, ↓30.42% MAE, ↑2.05% R ²
Ho et al. [2]	2021	Collaborative Network	Integrates candlestick patterns with Twitter sentiment analysis using 1D and 2D CNN models	Achieved 75.38% accuracy (Apple stock); performance improves for longer prediction horizons
Zeng et al. [3]	2023	RVGWI	Applies rising-trend visibility graph with Weisfeiler-Lehman (WL) kernel to capture structural patterns in time series	Outperforms traditional rule-based models; effectively captures topological relationships
Zatwardicki et al. [4]	2024	RSHR	Introduces rolling-window strategy for evaluating trading strategies over time	Reduces recency bias and enables comprehensive, robust strategy evaluation
Moustafa et al. [5]	2021	IoB Ecosystem	Combines real-time financial data with BERT-based sentiment analysis, LSTM prediction, and explainable AI (XAI)	High performance: RMSE 0.0312, 95.24% accuracy; portfolio growth from \$15K to \$21.8K

Existing Methodology

- Stock market analysis is mainly done using basic charts and simple methods
- Line charts show only closing prices and do not give full price details
- Basic statistical methods do not clearly explain volatility or risk
- Candlestick charts provide better information but are not combined with deeper analysis
- Most systems focus only on trends and ignore price changes (volatility)
- No proper visualization of return distribution, making risk hard to understand
- Many systems are not user-friendly, especially for beginners
- Overall, existing systems do not give a complete understanding of price behavior and volatility

Drawbacks of existing system

- Line charts show only closing prices and miss other important values
- Because of this, full price movement is not clear
- Systems mainly focus on whether prices go up or down
- They do not explain how much prices change (volatility)
- Too much numerical data makes it hard to understand
- There is no proper use of visual tools to explain data clearly
- Return analysis is not used, so price changes are not explained properly
- Candlestick charts are used but not combined with other analysis methods
- Risk and market uncertainty are not clearly shown
- Sudden price changes are difficult to identify
- Many systems are not easy to use, especially for beginners

Software Details

- The project is developed using Python for data analysis and visualization
- Jupyter Notebook is used for writing and testing code step by step
- Streamlit is used to create an interactive dashboard
- Pandas is used for data loading and preprocessing
- NumPy is used for numerical calculations
- Plotly is used for creating charts and visualizations

Research Gaps

- Most existing systems focus only on price trends and not on volatility or risk
- Visual analysis and statistical analysis are studied separately
- Important factors like return distribution and price changes are not properly analyzed
- There is no clear connection between price movement and numerical analysis
- Lack of interactive dashboards makes data exploration difficult
- No system combines visualization and statistical methods effectively
- Need for a system that converts complex data into clear visual insights

Proposed Methodology

- The system is designed to analyze stock market volatility using both visualization and statistical methods
- First, stock data (Date, Open, High, Low, Close, Volume) is collected
- The data is cleaned and prepared for accurate analysis
- Next, daily returns are calculated to understand price changes over time
- Candlestick charts are created to show price movements clearly
- They help in identifying trends, patterns, and reversals
- Return distribution is used to study how prices change
- It helps in understanding frequency, variation, and risk
- All results are combined into an interactive dashboard
- This allows users to explore data easily
- The system gives both visual and statistical insights together
- This makes volatility analysis simple and clear

Sample Code

Step 1: Import required libraries

```
import pandas as pd          # for handling data
import numpy as np           # for numerical operations
import plotly.express as px   # for advanced plots
import plotly.graph_objects as go  # for candlestick and heatmap
```

Step 2: Load the dataset

```
df = pd.read_csv("stock_market_data_large.csv")
```

Step 3: Convert Date column into datetime format

```
df['Date'] = pd.to_datetime(df['Date'])
```

Step 4: Calculate daily returns (for volatility analysis)

```
df['Return'] = df['Close'].pct_change()
```

Step 5: Create candlestick chart for price movement

```
fig1 = go.Figure(data=[go.Candlestick(
    x=df['Date'],
    open=df['Open'],
    high=df['High'],
    low=df['Low'],
    close=df['Close']
)])
fig1.show()
```

Step 6: Create 3D scatter plot (RSI vs Volume vs Close)

```
fig2 = px.scatter_3d(df, x='RSI', y='Volume', z='Close', color='Target')
fig2.show()
```

Step 7: Create correlation heatmap between features

```
corr = df[['Open','High','Low','Close','Volume','RSI','MACD','Sentiment']].corr()
fig3 = go.Figure(data=go.Heatmap(z=corr))
fig3.show()
```

Packages / Libraries

Data Processing :

- Pandas (pd)
- ➤ Used for loading, cleaning, and managing the dataset
- NumPy (np)
- ➤ Used for numerical calculations and handling large data efficiently

Visualization:

- Plotly Express (px)
- ➤ Used for advanced visualizations like 3D plots and animated graphs
- Plotly Graph Objects (go)
- ➤ Used for detailed charts like candlestick charts and heatmaps

Data Sets

- The project uses a stock market dataset with more than 13,000 daily records
- Each record includes Date, Open, High, Low, Close, and Volume
- These features show complete price movement for each day
- The dataset also includes indicators like RSI, MACD, Sentiment, and Target
- These help in understanding trends and market behavior
- The data is used to create candlestick charts for visualizing price movements
- Daily returns are calculated to analyze price changes and volatility
- It also supports visualizations like heatmaps and distribution graphs
- Overall, the dataset helps in studying trends, risk, and market behavior

Result Analysis

- Candlestick charts show price movements clearly and help identify trends (bullish and bearish)
- Stock prices show frequent fluctuations, indicating market volatility
- Daily returns show both positive and negative price changes
- Return distribution helps understand frequency, variation, and risk
- 3D plots show relationship between indicators like RSI, Volume, and Close price
- Correlation heatmap shows strong relation between price features
- Indicators like RSI and MACD show weaker relationships
- Density and heatmap visuals highlight periods of high activity and volatility
- Overall, combining visualization and analysis gives better understanding of trends and risk

Advantages and Application

Advantages

- Converts complex financial data into simple visual charts
- Makes it easy to understand price movements and patterns
- Helps quickly identify bullish and bearish trends
- Detects trend reversals and sudden price changes
- Improves risk analysis by showing gains and losses clearly
- Supports faster and better decision-making
- Uses data-driven insights instead of assumptions
- Helps identify overbought and oversold conditions
- Useful for both short-term and long-term analysis

Applications

- Stock Trading Platforms: Analyze live price movements
- Intraday Trading: Decide quick buy and sell points
- Risk Management: Track market crashes and volatility
- Algorithmic Trading: Automatic trading using live data
- Portfolio Management: Monitor and adjust investments
- Financial Analysis: Study impact of news on prices
- High-Frequency Trading: Detect fast price changes
- Market Comparison Tools: Compare multiple stocks

Conclusion

- Candlestick charts help identify short-term trends and price movements
- Return distribution helps understand long-term risk and returns
- Together, they simplify complex financial data
- Improve risk analysis and decision-making
- Combine visual and statistical methods for better insights
- Help investors make smarter and more stable investment decisions

Future Scope

- Use of AI & ML for better prediction of market trends
- Integration of real-time analytics for faster insights
- Development of smart candlestick charts with automatic pattern detection
- Improved return analysis using simulation models
- Use of big data to analyze multiple assets together
- Integration with sentiment analysis to understand market reactions
- Application in algorithmic trading for automated decisions
- Stronger risk management systems for better strategies
- Creation of smart dashboards that predict future movements
- Faster and more reliable data-driven decision-making tools

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